



NeuroSpace

Complete evaluation export

Job ID	2a396a88-fa10-4d99-8a3e-3b1b8d8c8cb2
Status	completed
Result timestamp	2026-05-20T22:46:18.545039
PDF generated	2026-05-20 23:51:49

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
manual_overrides.rnn_epochs	30
manual_overrides.qrnn_epochs	15
manual_overrides.batch_size	32
manual_overrides.rnn_batch_size	32
manual_overrides.qrnn_batch_size	16
num_classes	2
class_names	0, 1
qrnn_backend	qiskit
noise_enabled	True
noise_level	0.0100
mitigation_enabled	True
mitigation_runs	3

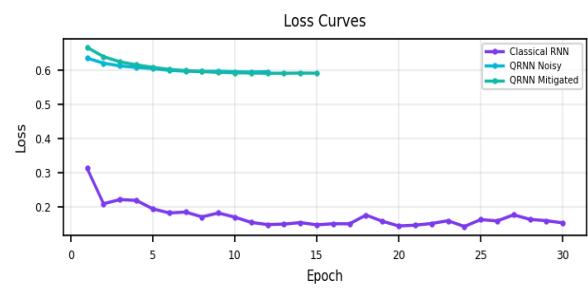
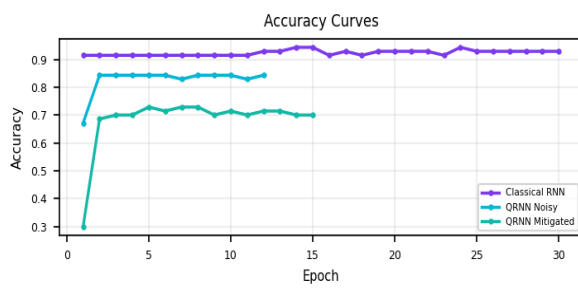
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	95.00%	82.14%	12.86%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	95.00%	95.03%	95.00%	95.01%	0.1527
QRNN Noisy	82.14%	83.43%	82.14%	80.78%	0.5957
QRNN Mitigated	77.14%	76.58%	77.14%	76.21%	0.5913

Learning Curves



Classical RNN

Metric	Value
Accuracy	95.00%
Precision	95.03%
Recall	95.00%
F1 Score	95.01%
Loss	0.1527
Train Loss	0.0346
Val Loss	0.1527
Train Accuracy	98.98%
Val Accuracy	92.86%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	3	45

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	96.70%	95.65%	96.17%	92.0000
1	91.84%	93.75%	92.78%	48.0000
macro avg	94.27%	94.70%	94.48%	140.0000
weighted avg	95.03%	95.00%	95.01%	140.0000
Accuracy			95.00%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.4199	0.3123	79.14%	91.43%
2	0.1294	0.2088	97.14%	91.43%
3	0.0945	0.2210	97.34%	91.43%
4	0.1050	0.2186	96.93%	91.43%
5	0.0807	0.1939	97.75%	91.43%
6	0.0890	0.1818	97.34%	91.43%
7	0.0783	0.1843	97.96%	91.43%
8	0.0616	0.1703	98.36%	91.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.0729	0.1816	97.96%	91.43%
10	0.0757	0.1696	97.96%	91.43%
11	0.0784	0.1540	97.55%	91.43%
12	0.0789	0.1478	97.75%	92.86%
13	0.0668	0.1490	97.96%	92.86%
14	0.0839	0.1535	97.75%	94.29%
15	0.0943	0.1469	96.93%	94.29%
16	0.0843	0.1502	97.55%	91.43%
17	0.0612	0.1500	98.16%	92.86%
18	0.0889	0.1755	97.96%	91.43%
19	0.0658	0.1574	98.16%	92.86%
20	0.0662	0.1437	98.36%	92.86%
21	0.0608	0.1459	98.36%	92.86%
22	0.0483	0.1509	98.36%	92.86%
23	0.0548	0.1590	98.16%	91.43%
24	0.0724	0.1421	97.96%	94.29%
25	0.0631	0.1623	97.75%	92.86%
26	0.0787	0.1584	96.73%	92.86%
27	0.0479	0.1762	98.77%	92.86%
28	0.0415	0.1629	98.57%	92.86%
29	0.0551	0.1590	98.36%	92.86%
30	0.0346	0.1527	98.98%	92.86%

QRNN Noisy

Metric	Value
Accuracy	82.14%
Precision	83.43%
Recall	82.14%
F1 Score	80.78%
Loss	0.5957
Train Loss	0.5929
Val Loss	0.5957
Train Accuracy	83.23%
Val Accuracy	84.29%
Train Size	489
Val Size	70
Test Size	140

QRNN Noisy Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	89	3
Class 1	22	26

QRNN Noisy Classification Report

Class	Precision	Recall	F1 Score	Support
0	80.18%	96.74%	87.68%	92.0000
1	89.66%	54.17%	67.53%	48.0000
macro avg	84.92%	75.45%	77.61%	140.0000
weighted avg	83.43%	82.14%	80.78%	140.0000
Accuracy			82.14%	

QRNN Noisy Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6636	0.6355	35.58%	67.14%
2	0.6433	0.6207	79.55%	84.29%
3	0.6331	0.6127	83.84%	84.29%
4	0.6253	0.6083	83.44%	84.29%
5	0.6194	0.6049	83.64%	84.29%
6	0.6133	0.5998	83.23%	84.29%
7	0.6071	0.5973	83.44%	82.86%
8	0.6071	0.5966	83.44%	84.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.6004	0.5969	83.44%	84.29%
10	0.5994	0.5955	83.44%	84.29%
11	0.5947	0.5946	83.23%	82.86%
12	0.5929	0.5957	83.23%	84.29%

QRNN Mitigated

Metric	Value
Accuracy	77.14%
Precision	76.58%
Recall	77.14%
F1 Score	76.21%
Loss	0.5913
Train Loss	0.5501
Val Loss	0.5913
Train Accuracy	80.98%
Val Accuracy	70.00%
Train Size	489
Val Size	70
Test Size	140

QRNN Mitigated Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	82	10
Class 1	22	26

QRNN Mitigated Classification Report

Class	Precision	Recall	F1 Score	Support
0	78.85%	89.13%	83.67%	92.0000
1	72.22%	54.17%	61.90%	48.0000
macro avg	75.53%	71.65%	72.79%	140.0000
weighted avg	76.58%	77.14%	76.21%	140.0000
Accuracy			77.14%	

QRNN Mitigated Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.6822	0.6671	32.92%	30.00%
2	0.6510	0.6397	56.44%	68.57%
3	0.6279	0.6247	78.32%	70.00%
4	0.6130	0.6157	79.96%	70.00%
5	0.6020	0.6085	79.75%	72.86%
6	0.5921	0.6025	79.96%	71.43%
7	0.5849	0.5988	80.98%	72.86%
8	0.5766	0.5962	81.39%	72.86%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
9	0.5711	0.5938	81.19%	70.00%
10	0.5671	0.5926	80.98%	71.43%
11	0.5647	0.5919	80.98%	70.00%
12	0.5593	0.5912	80.98%	71.43%
13	0.5573	0.5912	81.19%	71.43%
14	0.5521	0.5917	80.98%	70.00%
15	0.5501	0.5913	80.98%	70.00%